

LESSON 2-8 • SESSION 2 • PRACTICE • FORM A**Reaching a Target Mean**

Learning target – I can find the value a data set still needs in order to reach a target mean. • 6.SP.B.5c

REMEMBER

- **Target total = target mean × number of values.**
Count the missing value too.
- **Missing value = target total – sum so far.**
- If the missing value is more than the top score allowed, the goal is **not possible**.

WORKED EXAMPLE

Quiz scores are 6, 7 and 9. What fourth score gives a mean of 8?

Step 1 Count all scores, with the missing one: 4.

Step 2 Target total: $8 \times 4 = 32$. Sum so far: $6 + 7 + 9 = 22$.

Step 3 $32 - 22 = 10$. Check: $6 + 7 + 9 + 10 = 32$ and $32 \div 4 = 8$.

Answer The fourth score is 10.

WORDS target mean (*media objetivo*) • target total (*total objetivo*) • missing value (*valor faltante*)

PRACTICE

show your work • label units

1. Fill in each blank.

a. Mean 20, 4 values. Total = _____

b. Mean 8, 7 values. Total = _____

c. Need 72, have 57. Missing = _____

d. Need 120, have 96. Missing = _____

2. **Swim Meet.** A swimmer scored 27, 31, 24 and 30 points in four meets. She wants a mean of 29 points over five meets.

a. Target total: _____

b. Sum so far: _____

c. Meet five: _____

3. **Bowling Goal.** Ravi bowled 96, 104, 88 and 112. What score in game 5 gives a mean of 102?

Hint Turn the target mean into a target total first.

Show your work.

Score _____

4. **Missing Number.** Five numbers have a mean of 22. Four of them are 19, 25, 21 and 28. Find the fifth number.

Show your work.

Number _____

5. **One Perfect Quiz.** Dana's four quiz scores have a mean of 70. She says a 100 on quiz 5 makes her mean 100. Find her real new mean.

Hint Rebuild the total, then share it across 5 quizzes.

Show your work.

New mean _____

6. **Working Backward.** Six temperatures have a mean of 45°F. Five of them are 41, 39, 50, 47 and 44°F. Find the sixth temperature.

Hint Multiply by 6, the count that includes the missing value.

Show your work.

Answer _____ °F

- 7. Impossible Target.** Elias scored 70, 74 and 78 on three tests (top score 100). What score does he need for a mean of 90 over four tests? Is it possible?

Hint Compare the score needed with 100, the top score.

Show your work.

Score needed _____

- 8. Two Targets.** Three deliveries weighed 14, 20 and 17 kg. What must a fourth delivery weigh for each mean?

a. Mean of 18 kg: _____ kg

b. Mean of 20 kg: _____ kg

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Gideon's Work.** Gideon has a total of 104 points after four games. He wants a mean of 30 over five games, so he wrote $104 - 30 = 74$. Explain his mistake and find the score he needs.

TEST PRACTICE

choose the best answer

- 10.** Four values have a mean of 20. Their total is –

A 5

B 24

C 80

D 16

- 11.** Three scores total 51. What fourth score gives a mean of 15?

A 9

B 15

C 66

D 36

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

100 Q	76 G	33 T	110 A	74 Z	222 X	21 S	49 E	17 R	138 T
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#2c	#3	#4	#5	#6	#7	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____